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BrainNet: a fusion assisted novel optimal framework of residual blocks and stacked autoencoders for multimodal brain tumor classification

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A significant issue in computer-aided diagnosis (CAD) for medical applications is brain tumor classification. Radiologists could reliably detect tumors using machine learning algorithms without extensive surgery. However, a few important challenges arise, such as (i) the selection of the most important deep learning architecture for classification (ii) an expert in the field who can assess the output of deep learning models. These difficulties motivate us to propose an efficient and accurate system based on deep learning and evolutionary optimization for the classification of four types of brain modalities (t1 tumor, t1ce tumor, t2 tumor, and flair tumor) on a large-scale MRI database. Thus, a CNN architecture is modified based on domain knowledge and connected with an evolutionary optimization algorithm to select hyperparameters. In parallel, a Stack Encoder–Decoder network is designed with ten convolutional layers. The features of both models are extracted and optimized using an improved version of Grey Wolf with updated criteria of the Jaya algorithm. The improved version speeds up the learning process and improves the accuracy. Finally, the selected features are fused using a novel parallel pooling approach that is classified using machine learning and neural networks. Two datasets, BraTS2020 and BraTS2021, have been employed for the experimental tasks and obtained an improved average accuracy of 98% and a maximum single-classifier accuracy of 99%. Comparison is also conducted with several classifiers, techniques, and neural nets; the proposed method achieved improved performance.

Keywords Brain tumor, Explinable AI, MRI, Residual blocks, Stack encoders, Feature selection, Fusion, Classification

An abnormal cell growth developed in the brain is a brain tumor[1]. The microscopic organization of the tissues and distinctive molecular properties of brain tumors serve as classification criteria[2]. The medical team can better forecast a brain tumor's behavior, choose the best course of therapy, and select the potential candidates for clinical trials by knowing the kind and grade of the tumor[3]. According to the World Health Organization (WHO), there are almost 120 types of brain tumors. Each category is further subdivided into two general categories: benign and malignant. According to WHO, 700,000 Americans are surviving with primary brain tumors, whereas 71% have benign tumors and the rest have malignant tumors[4]. The benign tumor is considered a non-cancerous tumor. This kind of tumor does not spread to other parts of the body. They remain in their original location and are usually not problematic. They often have clear borders and a modest rate of growth. Usually, this kind of tumor is small, slow-growing, noninvasive, well-differentiated, and confined[5]. The malignant types of tumors are considered as

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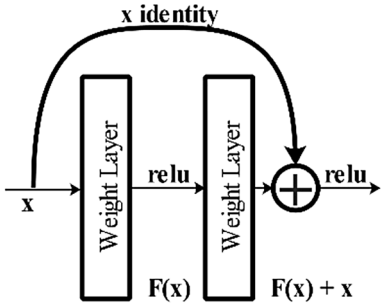

Figure 3. ResNet-50 skip connection.

Modification and Training In the first modification step, we added three new residual blocks, consisting of 4 convolutional layers. Each convolutional layer has 1 × 1 and 3 × 3 convolutional filters, and the stride is 1. After the first block, one convolutional layer of filter size 3 × 3 and stride 1 is added. The ReLu activation layer is added after each convolutional layer to resolve the nonlinearity issue. After adding three blocks, a global average pooling layer is added, as show in Fig. 4. In the end, a fully connected layer, softmax layer, and classification layers have been added and trained on the model on the selected dataset. The training process is conducted through transfer learning[45] without freezing any layer. The parameters of the new fully connected layer and overall parameters are defined in the Table 1 below. The hyperparameters have been selected based on the Jaya optimization

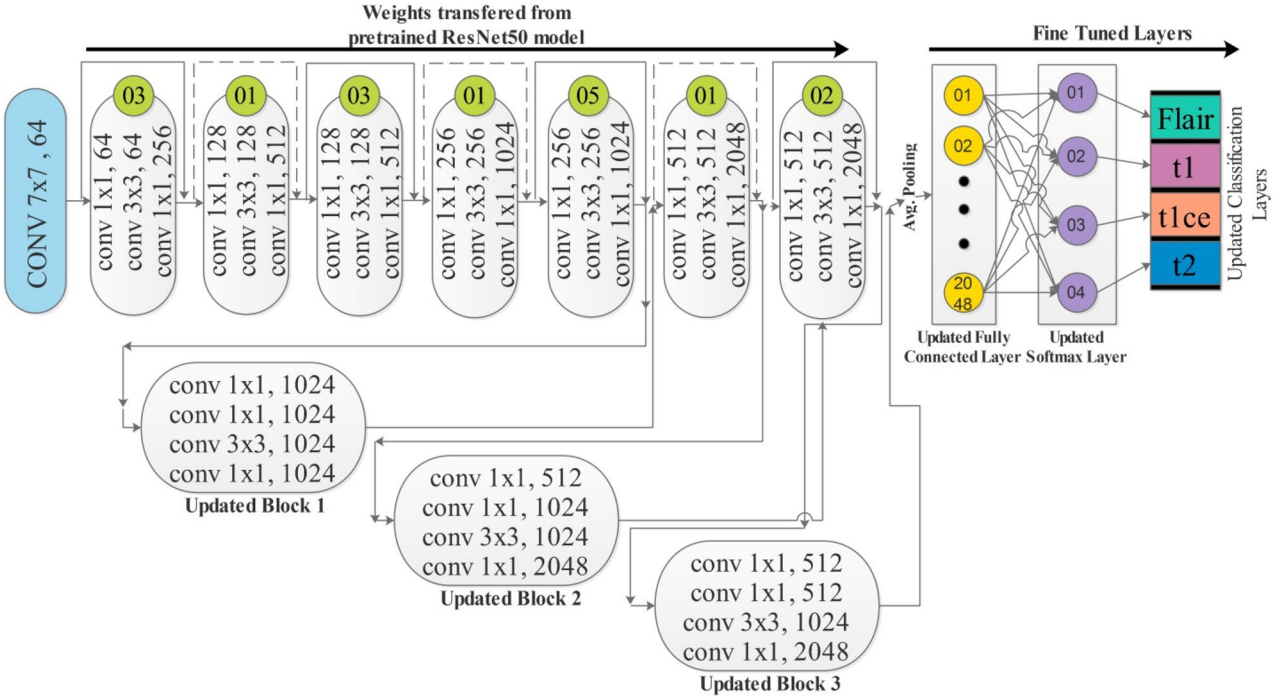

Figure 4. Proposed improved version of ResNet-50 CNN architecture.

Layer name	Fully connected layer
WeightLearnRateFactor	10
BiasLearnRateFactor	10
Optimizer	Stochastic gradient descent momentum (sgdm)
Maximum number of epochs	100
Minimum batch size	256
Initial learner rate	0.001

Table 1. Parameters of trained modified ResNet-50 CNN architecture.

algorithm[46]. The purpose of this step is to fully automate the system for the initialization of hyperparameters without involving human experience.

Novelty 2: stack auto-encoder network

A Stacked Auto-Encoder is a deep learning neural network architecture comprising multiple levels of encoders, which is succeeded by several layers of decoders[47]. The primary objective of this architecture is to learn a compressed representation of the input data that can be utilized for tasks like dimensionality reduction or representation learning. The Encoder component of the network transforms the input data into a bottleneck layer or latent representation, representing the data in lower dimensions. The decoder component then maps the lower-dimensional representation to the original input data space. The network is trained to minimize the difference between the decoder output and the original input data, ensuring the decoder output is as close to the input data. The loss function is used to minimize the difference.

An Auto-encoder with several layers (N), and the input data with (D) dimensions, then encoder, decoder, and loss function can be calculated using the below equations.

Encoder Consider each layer, the encoding function can be defined as:

$$en_i = act_i\left(wt_i * op_i - 1 + bias_i\right) \tag{4}$$

where output en_i is encoding function, act_i is the activation function, wt_i is the weight matrix, $op_i - 1$ is an output of the previous layer, and $bias_i$ is the bias vector of layer i.

Decoding The decoding function of layer i can be defined as:

$$de_i = dact_i(dwt_i * den_i + dbias_i) \tag{5}$$

where de_i is the output of the decoding function, $dact_i$ activation function, den_i is the input from previous encoding layer, dwt_i is weight matrix and $dbias_i$ is the bias vector, of decoding layer i.

Loss function The mean squared error between the input data and the output of the terminating decoding layer is acts as a loss function and can be denoted with following equation:

$$L = \frac{1}{(2N)} * \|\mathrm{I} - \mathrm{dOUT}\|^2 \tag{6}$$

In the equation, I denotes the input, dOUT is the output of the terminating layer, and the squared Euclidean norm is represented by $\|\cdot\|^2$. After the assignment of these functions, training is performed, and a trained model is later utilized for the feature extraction process. An overall architecture of proposed Stacked Autoencoders is showing in Fig. 5.

Feature extraction

After training two deep learning models named Modified ResNet-50 and Stack Auto Encoder–Decoder, features are extracted from the deeper layer. In the Modified ResNet-50 architecture, the global average pooling layer is employed for the feature extraction, whereas the last decoder convolutional layer is opted for the encoder-decoder network feature extraction. Both feature vectors are defined as follows:

$$\psi_{fv1} = \varrho\left(\vartheta_k, gap\right) \tag{7}$$

$$\psi_{fv2} = \Upsilon\left(\vartheta_k, Dconv\right) \tag{8}$$

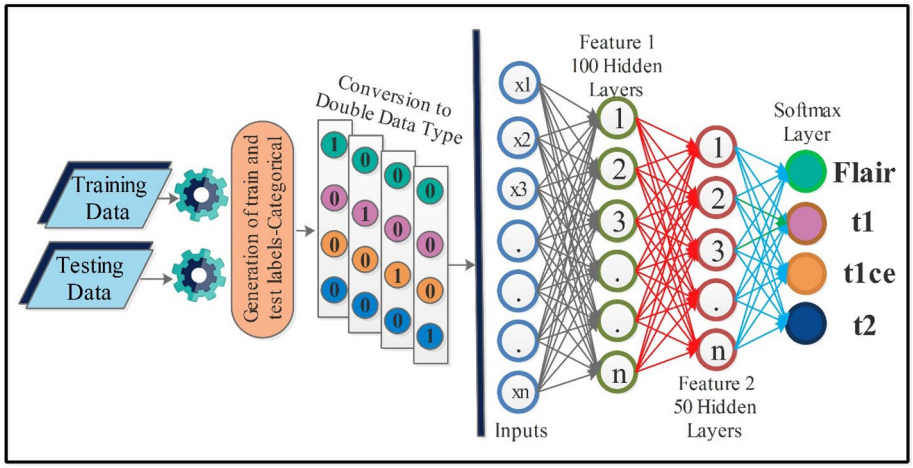

Figure 5. Proposed Stack Autoencoder network for features extraction.

where, ϑ_k denotes the data samples that are utilized for the feature extraction, ψ_{fv1} denotes the Modified Resnet-50 architecture, ψ_{fv2} denotes the Stack Auto Encoder-Decoder network, and *Dconv* is the decoder convolutional layer. The size of both feature vectors is $N \times 1024$ and $N \times 1236$, respectively.

Novelty 3: proposed feature fusion and selection

Feature fusion is the process of combining information of more than 1 sources into a single source for the improved information[48]. In this work, a novel parallel pooling concept-based approach is proposed for the fusion of both feature vectors. Consider we have two feature vectors ψ_{fv1} and ψ_{fv2} of dimensions $N \times 1024$ and $N \times 1236$. As per the parallel fusion definition, both vectors should have an equal length; therefore, we performed mean padding ψ_{fv1}. After that, a pooling operation is performed of filter size 2×2 and stride 1.

$$Fusion(k1) = \widetilde{\Delta}\left(\psi_{fv1}, stride, filter\right) \tag{9}$$

$$Fusion(k2) = \widetilde{\Delta}\left(\psi_{fv2}, stride, filter\right) \tag{10}$$

$$Fusion(\widetilde{k}) = \begin{pmatrix} Fusion(k1) \\ Fusion(k2) \end{pmatrix} \tag{11}$$

where $Fusion(k1)$ and $Fusion(k2)$ denote the resultant pooling vectors of dimensional 512 and 618, after that, both vectors are combined using Eq. (11) and obtained feature vectors of dimensional $N \times 1130$. The fused vector is optimized using an improved version of Grey Wolf Optimization algorithm with update criteria of Jaya algorithm.

Feature selection is selecting the best features later utilized for the classification[49]. This step aims to improve the accuracy and reduce the computational cost. The original extracted features consist of redundant and irrelevant information impacting classification accuracy and computational time. In addition, cleaning irrelevant and redundant information is important to obtain the classification results quickly. To address these challenges, a hybrid technique is proposed based on the initial selection of Grey Wolf Optimization and later again selected through an update equation.

The GWO is a nature-inspired algorithm presented by Mirjalili et al.[50] that impersonates grey wolves' hunting (optimized) behavior. Consider we have two feature vectors. ψ_{fv1} and ψ_{fv2} of dimensions $N \times 1024$ and $N \times 1236$. We initialize the number of populations 50 and iterations 200. The GWO is work in the following five steps:

Social hierarchy The hunting or dominance hierarchy involves alpha (α), beta (β), and delta (δ). The omega (ω) member follows the dominance of all.

Encircling prey To hunt, grey wolves encircle the prey, the encircling behavior can be modeled mathematically as follows:

$$\overrightarrow{L} = \overrightarrow{B} . \overrightarrow{Z}_p(t) - \overrightarrow{Z}(t) \tag{12}$$

$$\overrightarrow{Z}(t+1) = \overrightarrow{Z}_p(t) - \overrightarrow{M} . \overrightarrow{L} \tag{13}$$

where current iteration is indicated by m. The coefficient vectors are denoted by $\overrightarrow{M}$ and $\overrightarrow{B}$. The position vector of the prey is $\overrightarrow{Z}_p$. The position vector of the prey is indicated by $\overrightarrow{Z}$. The coefficient vectors are denoted by $\overrightarrow{M}$ and $\overrightarrow{B}$ can be calculated as follows:

$$\overrightarrow{M} = 2\overrightarrow{g} . \overrightarrow{h_1} - \overrightarrow{g} \tag{14}$$

$$\overrightarrow{M} = 2 . \overrightarrow{h_2} \tag{15}$$

An iteration-based linear decrement is done from 2 to 0 in the components of $\overrightarrow{g}$. The $\overrightarrow{h_1}$ and $\overrightarrow{h_2}$ are two vectors with the range of [0, 1].

Hunting Grey wolves hunt their prey precisely, and it is usually guided by alpha (α). The participation of beta (β) and delta (δ) is occasion-based. In a real scenario, the location of the prey always changes. In order to mathematically represent the problem space. We assume that the location of the prey is known by alpha (the Best Solution), beta, and delta. So, the first three best solutions obtained so far are saved and on the basis of this, we oblige the other search agents to update their position. Mathematical notation is given as follows: -

$$\overrightarrow{L_\alpha} = \overrightarrow{B_1} . \overrightarrow{Z}_\alpha - \overrightarrow{Z}, \overrightarrow{L_\beta} = \overrightarrow{B_2} . \overrightarrow{Z}_\beta - \overrightarrow{Z}, \overrightarrow{L_\delta} = \overrightarrow{B_3} . \overrightarrow{Z}_\delta - \overrightarrow{Z} \tag{16}$$

$$\overrightarrow{Z}_1 = \overrightarrow{Z}_\alpha - \overrightarrow{M}_1 . \left(\overrightarrow{L}_\alpha\right), \overrightarrow{Z}_2 = \overrightarrow{Z}_\beta - \overrightarrow{M}_2 . \left(\overrightarrow{L}_\beta\right), \overrightarrow{Z}_3 = \overrightarrow{Z}_\delta - \overrightarrow{M}_3 . (\overrightarrow{L}_\delta) \tag{17}$$

$$\overrightarrow{Z}(m+1) = \frac{\overrightarrow{Z}_1 + \overrightarrow{Z}_2 + \overrightarrow{Z}_3}{3} \tag{18}$$

Attacking prey As the prey stops moving, the value of $\overrightarrow{g}$ decreases. The variability range of $\overrightarrow{M}$ is also decreased by $\overrightarrow{g}$. In other words, we can say that vector $\overrightarrow{M}$ has an interval $[-2g, 2g]$, and the value of g iteratively decreased from 2 to 0. When Randomized values of vector $\overrightarrow{M}$ reaches between the interval $[-1, 1]$, the new position of the searching agent can be any between the current position and the prey's position. In this position, the wolves usually chose to attack the prey.

Search for prey To search for prey, grey wolves diverge and converge to attack the prey. Searching for prey usually involves alpha, beta, and delta members. The divergence mechanism can be mathematically modeled by selecting the random values of the vector $\overrightarrow{M}$ greater than 1 or less than − 1. The grey wolf algorithm searches globally by keeping in view the divergence mechanism. The vector $\overrightarrow{B}$ can also help in divergence. It has a range of [0, 2] random values. It emphasizes avoiding the problem of local optima and stagnation during the final iterations of the exploration stage.

The fitness is computed in each iteration and the best solution. The extreme learning machine (ELM)[51] classifier is employed as a fitness function in this work that computes the accuracy of each solution after each iteration. After all iterations, the best-selected solution is further optimized using an update function of Jaya optimization[52]. Mathematically, it is defined as follows:

$$X\prime_{l,m,j} = X_{l,m,j} + r_{1,l,j}\left(X_{l,best,j} - \left|X_{l,m,j}\right|\right) - r_{2,l,j}(X_{l,worst,j} - \left|X_{l,m,j}\right|) \tag{19}$$

In the equation $X_{l,best,j}$ is the value of the variable l for the best solution candidate, and $X_{l,worst,j}$ is the value of variable l for the worst solution candidate. The variables $r_{1,l,j}$ and $r_{2,l,j}$ are random variables that are generated between [0, 1] range for l^{th} variable and in the j^{th} iteration. The equation $+r_{1,l,j}\left(X_{l,best,j} - \left|X_{l,m,j}\right|\right)$ moves the overall solution closer to the best solution, while the equation $-r_{2,l,j}(X_{l,worst,j} - \left|X_{l,m,j}\right|)$ avoids the solution to be closer to the worst solution. The resultant value $X\prime_{l,m,j}$ is accepted if it provides us with better function values. The accepted resultant values are then saved in some variable and become the next iteration's input. In this paper, 100 iterations were selected for this second phase. The fitness value is computed in each iteration and the best solution ψ_{fv1}^{sl} is selected after all iterations. The best selected solution is finally passed to neural networks and machine learning algorithms. A brief working of the entire proposed method is described under Algorithm 1.

Algorithm 1

Input← *MRI Scans of BraTS*20 *and BraTS*21

Output← Final Prediction and Numerical Results such as

- T1, T1CE, T2, and Flair

Step 1: Normalized Datasets

Step 2: Modified ResNet-50 CNN Model

Step 3: Training of Step 2 CNN Model

Step 4: Designed Stack Auto-Encoder Network

Step 5: Training of Step 4 Network

Step 6: Feature Extraction

- ψ_{fv1} and ψ_{fv2}

Step 7: Proposed feature fusion using Eq. 10-12

- $Fusion(\tilde{k}) \leftarrow Fused\ feature\ vector$

Step 8: Best Feature Selection using Eq. 13-20

Step 9: Second phase selection using Eq. 20.

Step 10: Obtained a best solution feature vector ψ_{fv1}^{sl}

Step 11: Classify step 10 features using Neural Networks

Proposed results and discussion

Experimental setup

To perform experiments using improved version of ResNet-50, we have selected stochastic gradient descent with momentum (SGDM) to accelerate the gradient vectors for faster convergence at convolutional layers. We have used the k-fold cross-validation and selected the value of k = 10 for the model's training accuracy. We have divided the whole dataset into training and testing by the ratio of 50:50. During the algorithms learning process, we select a subset called a mini batch size and set its value to 256. Ultimately, we set the initial learner rate equal to 0.0001. On the other hand, while performing experiments with a Stacked Autoencoder, we have used two stacks of autoencoders. In the first stack, we selected hidden size = 100, the maximum number of epochs to 5, L2WeightRegularization = 0.004, SparsityRegularization = 4, and SparsityProportion = 0.15. In the second stack, we have used selected hidden size = 50, max number of epochs to 5, L2WeightRegularization = 0.002, SparsityRegularization = four, and SparsityProportion = 0.1. Moreover, the experiments were performed on MATLAB R2022a on a computer with 16 GB of RAM and a 12 GB Graphics Processing Unit (GPU) of RTX3060.

Datasets and performance measures

We have used MICCAI BraTS 2020 and MICCAI BraTS 2021 brain tumor MRI datasets for our experiments. Both of these datasets are available publically for research purposes. We use different performance measures to evaluate our methodology's experimental results on these datasets. Performance measures in machine learning are metrics used to evaluate how well a model or algorithm can predict outcomes based on the available data. The selected measures are sensitivity rate, false negative rate (FNR), precision rate, and area under the Curve of each classifier. Additionally, we have recorded the accuracy and training time for each classifier. More information regarding these performance protocols can be found in Table 2.

MICCAI BraTS 2020 results

Results of the proposed framework have been computed in several steps to show the importance of each middle step. The improved ResNet-50 CNN architecture is selected in the first step, and classification is performed in the feature extraction phase. The results of this phase are given in Table 3. This table shows 99.70% best accuracy with a minimum noted time of 206.75 (s) for Medium Neural Networks. Figure 6 shows the confusion matrix of this step to verify the obtained performance measures. The diagonal values in this figure illustrated each modality class's correct prediction (number of true observation and true positive rate).

In the second step, we employed Stacked Autoencoders architecture and extracted features, as shown in Fig. 4. The results of this step are given in Table 4. In this table, 98.90% best accuracy is noted for Wide Neural Network with 98.85% sensitivity rate and 98.85% precision rate. The other classifiers, named Medium Neural Network, Narrow Neural Network, and Tri-Layered Neural Network, obtained an accuracy of 97.90%, 92.70%, and 92.60%, respectively. The minimum accuracy was 92.40%, which was obtained by the Bi-Layered Neural Network classifier. Whereas the minimum noted time is 209.20 (sec) for Tri-Layered Neural Network. Figure 7 shows the confusion matrix for this step to verify the obtained performance measures. The diagonal values in this figure illustrate the correct prediction.

To enhance the accuracy, a novel fusion technique is utilized to fuse the information of both proposed models. The results of this step are given in Table 5. In this table, 99.90% best accuracy is noted for Wide Neural Network with 99.90% sensitivity rate and 99.80% precision rate. Whereas the minimum noted time is 249.38 (s) for Bi-Layered Neural Network. Figure 8 shows the confusion matrix for this step to verify the obtained performance measures. It is noted that after this step, the accuracy is improved by 0.2% and 1% for the improved ResNet-50

Name	Accuracy (%)	Time	Sensitivity rate (%)	False negative rate (%)	Precision rate (%)	Area under curve
Performance measure	$\frac{TP+TN}{TP+TN+FP+FN}$	*Seconds*	$\frac{TP}{TP+FP}$	$\frac{FN}{TP+FP}$	$\frac{TP}{TP+FN}$	$\int_a^b f(x)dx$

Table 2. Performance measures used to validate the proposed methodology. Where *TP* True Positive, *TN* True Negative, *FP* False Positive, *FN* False Negative.

Classifier	Accuracy (%)	Time (s)	Sensitivity rate (%)	False negative rate (%)	Precision rate (%)	Area under curve (%)
WNN	**99.70**	277.30	**99.68**	**0.32**	**99.70**	**1.00**
Medium NN	99.60	236.82	99.70	0.30	99.70	1.00
Bi-LayerNN	99.64	358.60	99.68	0.32	99.65	1.00
Tri-Layered NN	99.58	**206.75**	99.68	0.32	99.68	1.00
Narrow NN	99.60	233.20	99.65	0.35	99.65	1.00

Table 3. Classification results of the improved version of ResNet-50 CNN architecture. Significant values are in bold.

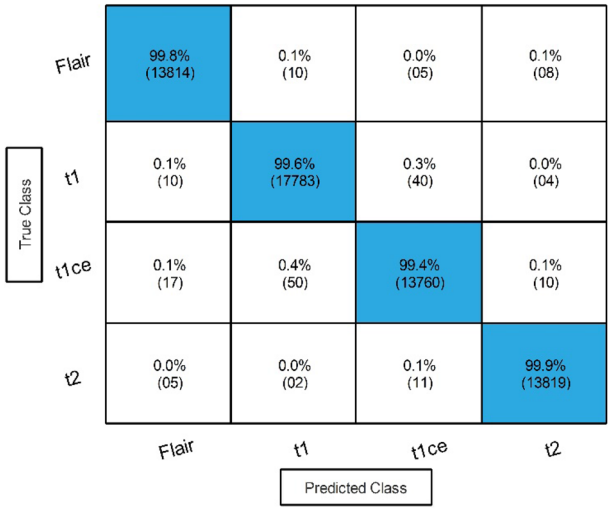

Figure 6. Confusion matrix of Tri-layered Neural Network for an improved version of ResNet-50 CNN architecture.

Classifier	Accuracy (%)	Time (s)	Sensitivity rate (%)	False negative rate (%)	Precision rate (%)	Area under curve (%)
WNN	**98.90**	303.70	**98.85**	**1.15**	**98.85**	**1.00**
Medium NN	97.90	232.83	97.90	2.10	97.88	1.00
Bi-LayerNN	92.70	223.5	92.75	7.25	92.77	0.985
Tri-Layered NN	92.60	237.30	92.60	7.40	92.63	0.988
Narrow NN	92.40	**209.20**	92.40	7.60	92.43	0.985

Table 4. Classification results by proposed Stacked Autoencoders model. Significant values are in bold.

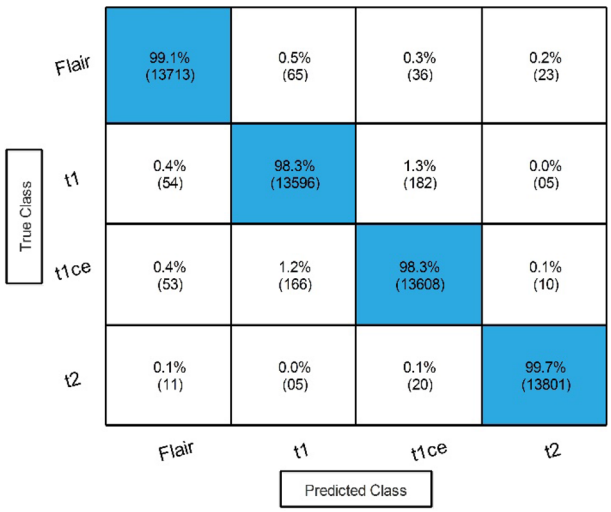

Figure 7. Confusion matrix of proposed stacked autoencoders.

and proposed Stacked Autoencoders model, respectively. However, an increase occurs in the computational time after the fusion process. An optimization technique is proposed in the next step to address this challenge.

In the final step, the proposed optimization algorithm is applied, and the best features are selected. The results of the proposed optimization step are given in Table 6. In this table, 99.80% best accuracy is noted for WNN, with a 99.75% sensitivity rate and 99.78% precision rate. The MNN obtained the second-highest accuracy of 99.70%, while the Bi-Layered Neural Network, Tri-Layered NN, and Narrow NN classifier obtained the minimum accuracy of 99.60%. Whereas the minimum noted time is 32.653 (sec) for Medium Neural Networks. Figure 9 shows

Classifier	Accuracy (%)	Time (s)	Sensitivity rate (%)	False negative rate (%)	Precision rate (%)	Area under curve (%)
WNN	**99.90**	**338.42**	**99.90**	**0.10**	**99.80**	**1.00**
Medium NN	99.82	**303.59**	99.88	0.12	99.80	1.00
Bi-LayerNN	99.80	389.38	99.88	0.12	99.80	1.00
Tri-Layered NN	99.82	**304.8**	99.88	0.12	99.80	1.00
Narrow NN	99.78	**363.24**	99.80	0.12	99.80	1.00

Table 5. Classification results after the fusion of improved ResNet-50 and proposed Stacked Autoencoder features. Significant values are in bold.

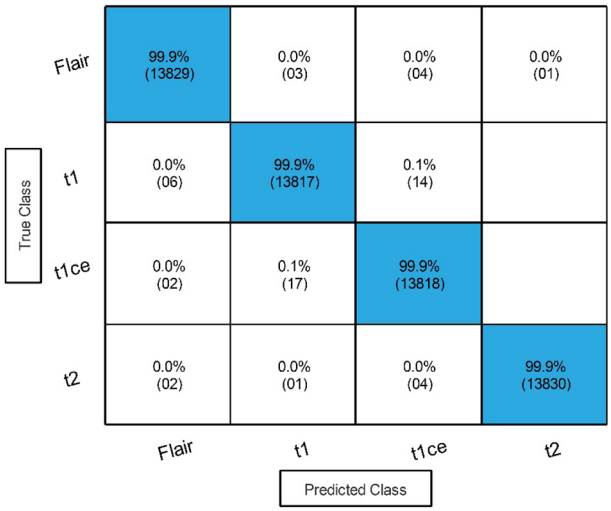

Figure 8. Confusion matrix of wide neural network using proposed fusion technique.

Classifier	Accuracy (%)	Time (s)	Sensitivity rate (%)	False negative rate (%)	Precision rate (%)	Area under curve (%)
WNN	**99.80**	53.419	**99.75**	**0.25**	**99.78**	**1.00**
Medium NN	99.70	**32.653**	99.70	0.30	99.70	1.00
Bi-LayerNN	99.60	51.435	99.63	0.37	99.63	1.00
Tri-Layered NN	99.60	85.33	99.63	0.37	99.63	1.00
Narrow NN	99.60	37.516	99.60	0.40	99.63	1.00

Table 6. Classification results of the proposed optimization technique using BraTS2020. Significant values are in bold.

the confusion matrix for this step to verify the obtained performance measures. This final step has improved the time complexity many folds than that of previous steps. The time complexity is decreased to approximately 87% in the third step, 98% in the second step, and approximately 95% in the first step. There is a negligible decrease (i.e., 0.1%) in accuracy in the third step due to dropping a few features during the selection step. However, this minor decrease is tolerable by keeping in view the improvement in the computational time. Figure 10 shows a detailed visual comparison in terms of time among all middle steps. In this figure, it is clearly observed that the time is significantly reduced after employing the proposed optimization technique.

MICCAI BraTS 2021 results

The same procedure of experiments was also carried out on the second dataset. The first step is performed for an improved version of ResNet-50, as shown in Fig. 4. Features are extracted from the global average pooling layer, and classification results are described in Table 7. The result describes that 99.90% best accuracy is noted. In addition, the minimum noted time is 41.692 (s) for Medium NN.

In the second experiment, the proposed Stacked Autoencoders model is utilized, and features are extracted. The results of this step are described in Table 8. The result describes that 99.60% best accuracy is noted along with 99.58% sensitivity rate and 99.55% precision rate for the Wide Neural Network classifier. Moreover, an accuracy of 99.50%, 98.80%, 98.80%, and 98.60% was noted for Medium, Bi-Layered, Narrow, and Tri-Layered Neural

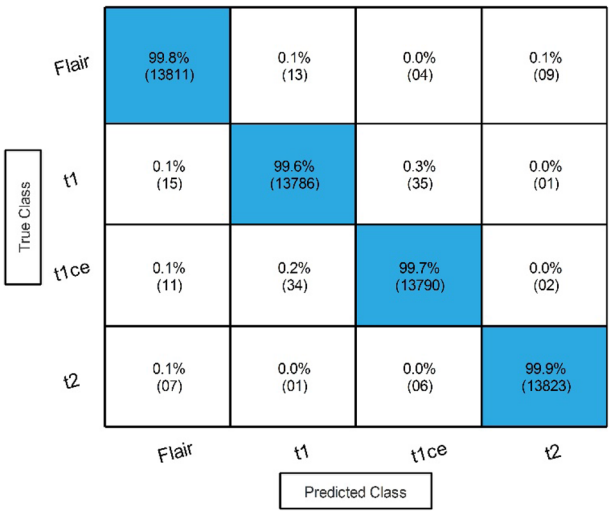

Figure 9. Confusion matrix of wide neural network using proposed optimization algorithm.

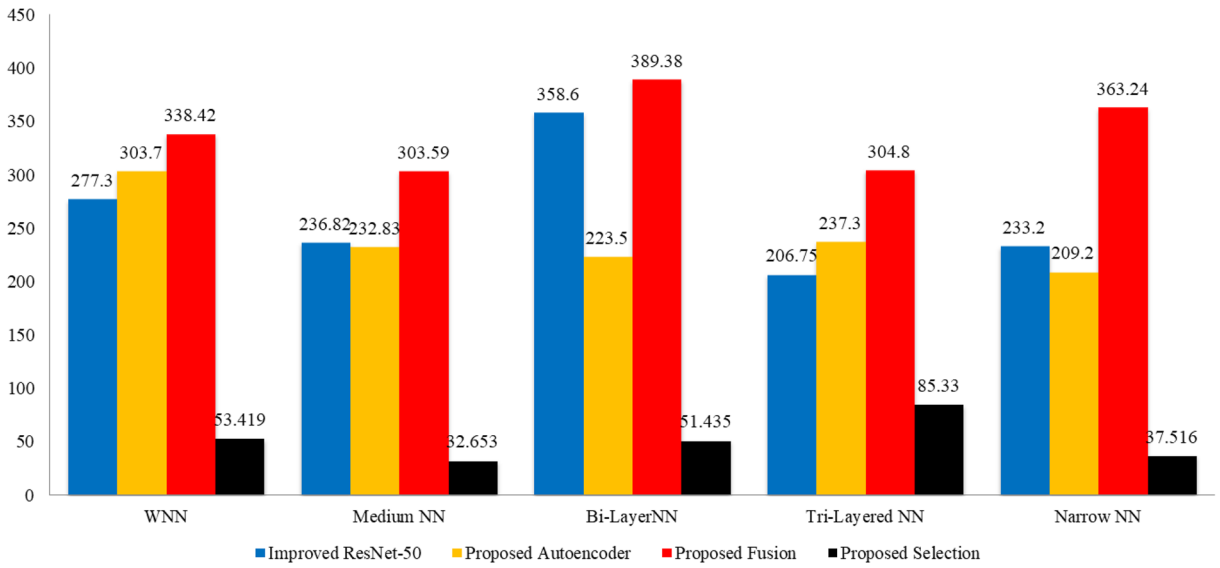

Figure 10. Computational time based comparison among all middle steps for BraTS2020 dataset.

Classifier	Accuracy (%)	Time (s)	Sensitivity rate (%)	False negative rate (%)	Precision rate (%)	Area under curve (%)
Medium NN	**99.90**	**41.692**	**99.87**	**0.12**	**99.87**	**1.00**
Narrow NN	99.90	46.518	99.87	0.13	99.87	1.00
Wide NN	99.90	60.325	99.87	0.13	99.85	1.00
Bi-Layered NN	99.80	45.644	99.83	0.17	99.85	1.00
Tri-Layered NN	99.80	62.655	99.78	0.22	99.80	1.00

Table 7. Proposed classification results of improved ResNet-50 model on BraTS2021. Significant values are in bold.

Networks, respectively. In addition, the minimum noted time is 67.558 (sec) for Medium NN. Compared to the previous experiment, the results show that the proposed Stacked Autoencoders obtained results in higher time (> 62%) than improved ResNet-50.

In the third step, a novel fusion technique is employed, and results are described in Table 9. The result describes that 100% best accuracy is noted along with a 99.98% sensitivity rate and 99.95% precision rate for the Narrow NN classifier. The other classifiers also obtained improved accuracy. The confusion matrix in Fig. 11

Classifier	Accuracy (%)	Time (s)	Sensitivity rate (%)	False negative rate (%)	Precision rate (%)	Area under curve (%)
Wide NN	**99.60**	88.419	**99.58**	**0.42**	**99.55**	**1.00**
Medium NN	99.50	**67.558**	99.48	0.52	99.50	1.00
Bi-Layered NN	98.80	129.64	98.80	1.20	98.83	1.00
Narrow NN	98.80	217.99	98.80	1.20	98.75	1.00
Tri-Layered NN	98.60	208.53	98.55	1.45	98.55	1.00

Table 8. Proposed classification results of stacked autoencoders model on BraTs2021 dataset. Significant values are in bold.

Classifier	Accuracy (%)	Time (s)	Sensitivity rate (%)	False negative rate (%)	Precision rate (%)	Area under curve (%)
Narrow NN	**100**	**156.692**	**99.98**	**0.02**	**99.95**	**1.00**
Medium NN	100	84.31	99.95	0.05	99.95	1.00
Wide NN	100	175.46	99.95	0.05	99.95	1.00
Bi-Layered NN	99.90	258.998	99.95	0.05	99.93	1.00
Tri-Layered NN	99.90	305.48	99.85	0.15	99.85	1.00

Table 9. Classification results after employing the proposed fusion technique. Significant values are in bold.

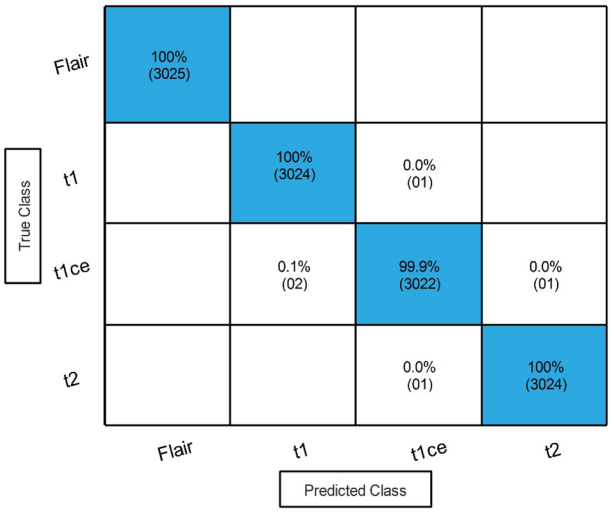

Figure 11. Confusion matrix of the best accuracy after employing the proposed fusion technique.

Classifier	Accuracy (%)	Time (s)	Sensitivity rate (%)	False negative rate (%)	Precision rate (%)	Area under curve (%)
Medium NN	**99.90**	**27.270**	**99.98**	**0.02**	**99.95**	**1.00**
Bi-Layered NN	99.90	30.232	99.95	0.05	99.95	1.00
Narrow NN	99.90	28.779	99.93	0.07	99.93	1.00
Wide NN	99.90	37.708	99.93	0.07	99.93	1.00
Tri-Layered NN	99.80	41.278	99.83	0.17	99.83	1.00

Table 10. Classification results after employing an improved version of Grey Wolf Optimization. Significant values are in bold.

verifies the obtained performance measures. The correct prediction is illustrated by diagonal values in the confusion matrix. It is further noted that the feature fusion step increased the performance in terms of accuracy and precision rate, whereas the computational time is also increased.

In the last step, an improved version of the Grey Wolf optimization technique is employed, and the results are described in Table 10. The result describes that 99.90% best accuracy is noted along with 99.98% sensitivity

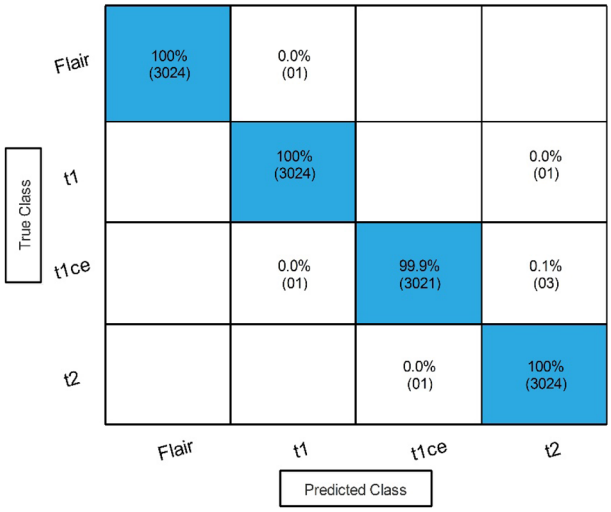

Figure 12. Confusion matrix of improved version of Grey Wolf Optimization on BraTS2021 dataset.

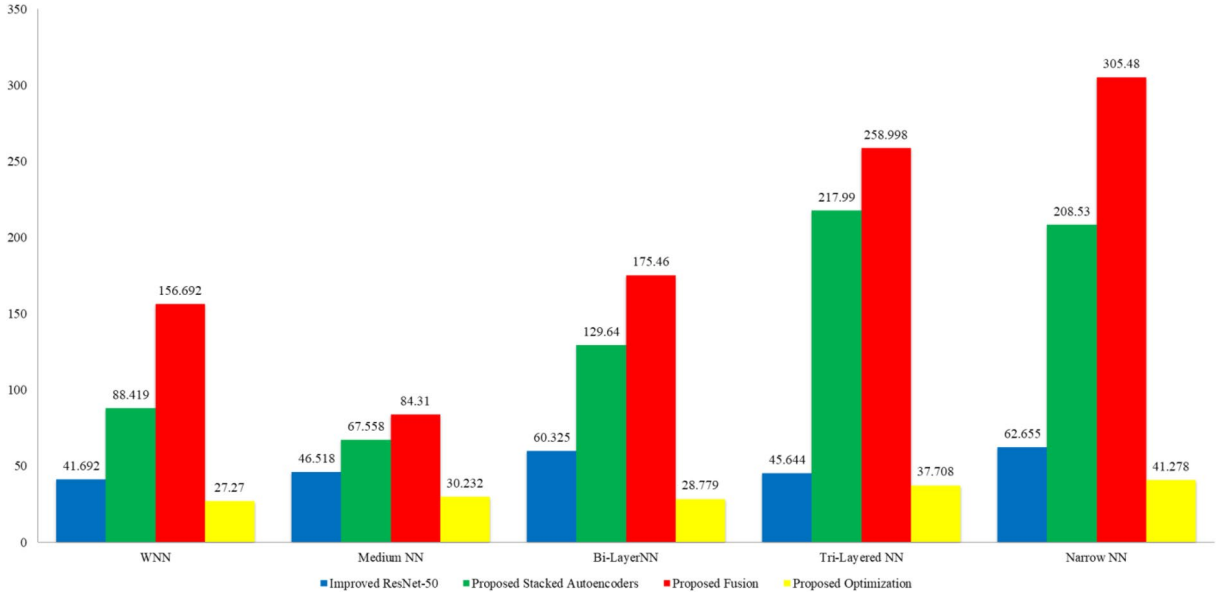

Figure 13. Comparison among all middle steps based on the computational time for BraTS2021 dataset.

rate and 99.95% precision rate for the Narrow NN classifier. Moreover, an accuracy of 99.90% was noted for Bi-Layered Narrow and Wide NN classifiers, while 99.80% accuracy was noted for Tri-Layered NN classifiers. In addition, the minimum noted time is 27.270 (s) for Medium NN. Figure 12 verifies the obtained performance measures. The correct prediction is illustrated by diagonal values in the confusion matrix. The accuracy of this phase is almost equal, but the computational time is significantly reduced, which is a strength of this work. A time comparison for this dataset among all middle steps is shown in Fig. 13. This figure shows the strength of the improved version of Grey Wolf Optimization.

Classifier	ResNet-50	Stacked autoencoders	Fusion process	Grey Wolf & Jaya optimizers
Wide neural network	99.70	98.90	99.90	99.80
Narrow neural network	99.70	92.70	99.90	99.60

Table 11. Classifiers with relevant accuracies for all experiments (BraTS2020).

Student's T-test based analysis

We have performed this test in order to validate our null hypothesis for both datasets, that is given as follows: h_0= There is a significant difference between the accuracies of selected classifiers. In order to perform our experiments, firstly, we have selected the two classifiers with corresponding accuracies in all experiments for a dataset of BraTS2020 dataset. One classifier which is performing with best accuracy among all experiments and the other classifier which is at the lowest level of performance among all the classifiers. The detail is given in Table 11.

The mean of the differences for all experiments are were calculated using following equations:

$$Difference(D) = \left|Accuracy(i) - Accuracy(j)\right| \tag{20}$$

$$Accuracy(i) = Accuracies\ of\ first\ classifier$$

$$Accuracy(j) = Accuracies\ of\ second\ classifier$$

$$Mean(\mu) = \frac{1}{N}\sum_{i=1}^{N}|D_i| \tag{21}$$

where N is the number of experiments and the noted mean value after this step is 1.6. After calculating the $Mean(\mu)$, the *standard deviation*(σ) is calculated by using following equation:

$$standard\ deviation(\sigma) = \sqrt{\frac{\left(\sum_{i=1}^{N}(D_i) - \mu\right)^2}{N-1}} \tag{22}$$

The resultant standard deviation value is 3.065 later used in the $T - Selection$ formula.

$$T - Selection = \frac{\sqrt{N} \times \mu}{\sigma} \tag{23}$$

The resultant value of this equation is 1.044. This value will be considered as a reference point to conduct the *Student's T-Test*. In our next step, we find the degree of freedom as $DegreeofFreedom\left(df\right) = n - 1$; hence, the df value is 3 and $pvalue = 0.05$[53]. In the t-table, the output value is $(-3.182, +3.182)$. As the t-selection value for this dataset is 1.044, based on the given below formulation, this hypothesis is rejected, and there is no significant difference in the accuracy of the two classifiers for all methods.

$$If\,(T - Selection >= -3.182\ and\ <= +3.182) \tag{24}$$

Similarly, this process is conducted for BraTS2021 and obtained the t-value of 1.5, df is 3, value of t in the table is $(-3.182, +3.182)$ at p-value 0.05. The t-value is not fall under this interval; hence, the hypothesis h_0 is rejected.

Grad-CAM based performance analysis of improved ResNet-50

The classification score depends upon the gradients; Grad-CAM utilizes it with respect to the obtained final convolutional feature map. This visualization method highlights the areas of the input image with a higher contribution to the classification score. The features are extracted from global average pooling layer. This method works using Eq. (25), which is given below:

$$a_k^c = \frac{1}{N}\sum_i\sum_j \frac{\partial_y^c}{\partial A_{i,j}^k} \tag{25}$$

where a_k^c are class scores of k features of the Global Average Pooling layer, N represents total pixels in a feature map, c depicts the class score, and y is considered as output. The whole expression $\partial A_{i,j}^k$ represents the convolution map. In the expression, i and j represent two dimensions, and A represents gradients. The output of this equation may be negative feature weights; therefore, a ReLu activation function is employed to resolve this issue.

$$\mathrm{M} = \mathrm{Relu}\left(\sum a_k^c.A^k\right) \tag{26}$$

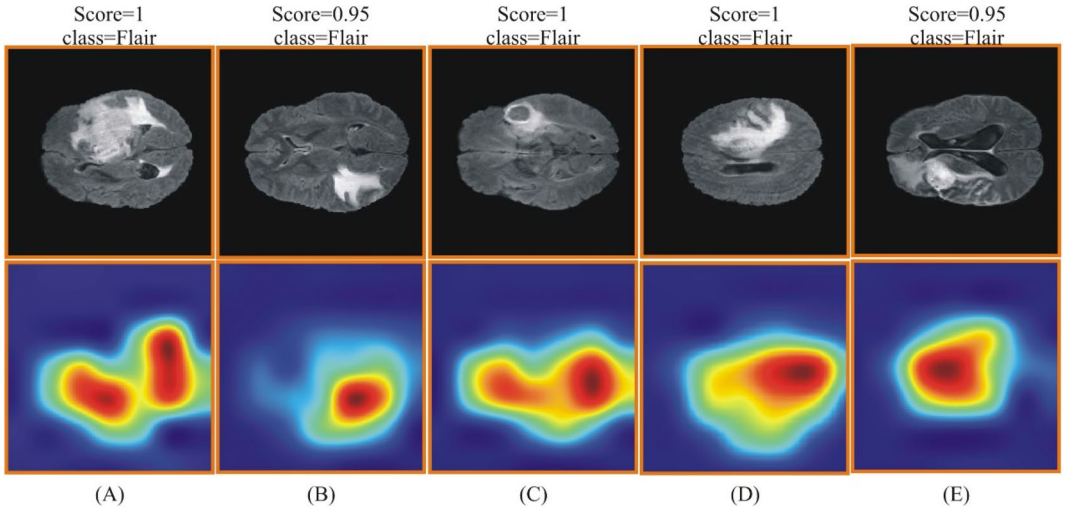

Figure 14. Grad-CAM based heat map for BraTS2020 dataset using improved ResNet-50. (**A**–**E**) showing the original and heat map images.

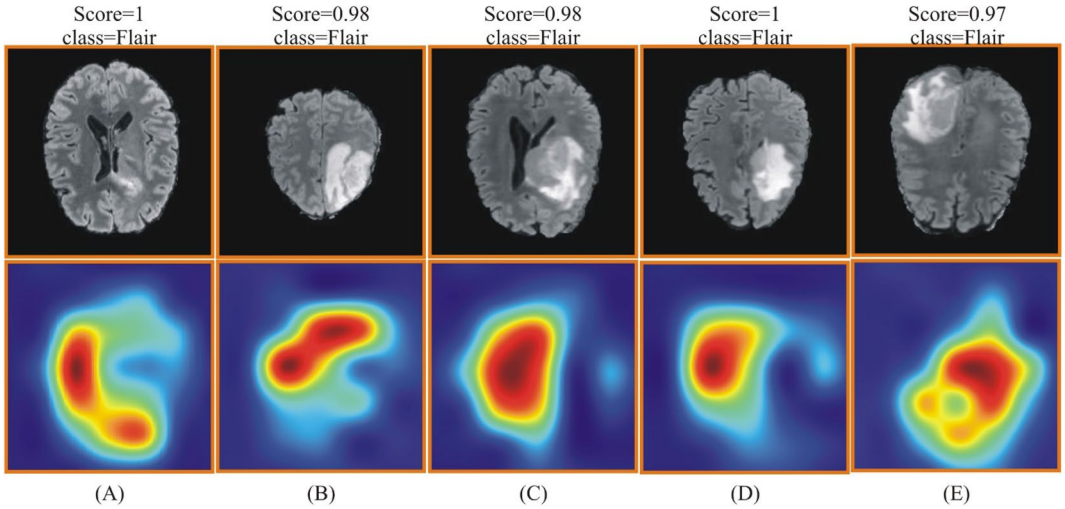

Figure 15. Grad-CAM based heat map visualization for BraTS 2021 dataset using improved ResNet-50. (**A**–**E**) showing the original and heat map images.

The visual results of this method are shown in Figs. 14 and 15. These figures show that the score and class output are returned against each image as a predicted label. The highlighted region is the most important region utilized for the corresponding label.

After the visualization, a short discussion is conducted based on graphical results. The discussion is conducted for three different scenarios: (i) Original ResNet-50 vs. Improved Version of ResNet-50; (ii) comparison of improved ResNet-50 with several other neural nets; and (iii) Original Grey Wolf vs. Improved version of Grey Wolf and with several other methods. Figure 16 shows the results of these scenarios. This figure clearly illustrates that the improved version accuracy is better than the original neural nets and optimization algorithms. Lastly, a comparison is also conducted with recent state-of-the-art techniques, as described in Table 12. This table shows that the proposed method obtained improved accuracy than the recent techniques on the selected datasets.

Conclusion

The research of medical image applications that employ deep learning and optimization strategies and adhere to the concept of effective natural images in practical applications has attracted more attention in recent years. MRI data has been frequently used with deep learning techniques for automatic brain tumor detection and classification. This paper provides a fully automated computerized multiclass brain tumor classification system with deep learning and optimization. In the proposed architecture, an improved ResNet-50 model is designed and automatically initializes the hyperparameters using the Jaya algorithm that is later employed to learn a model.

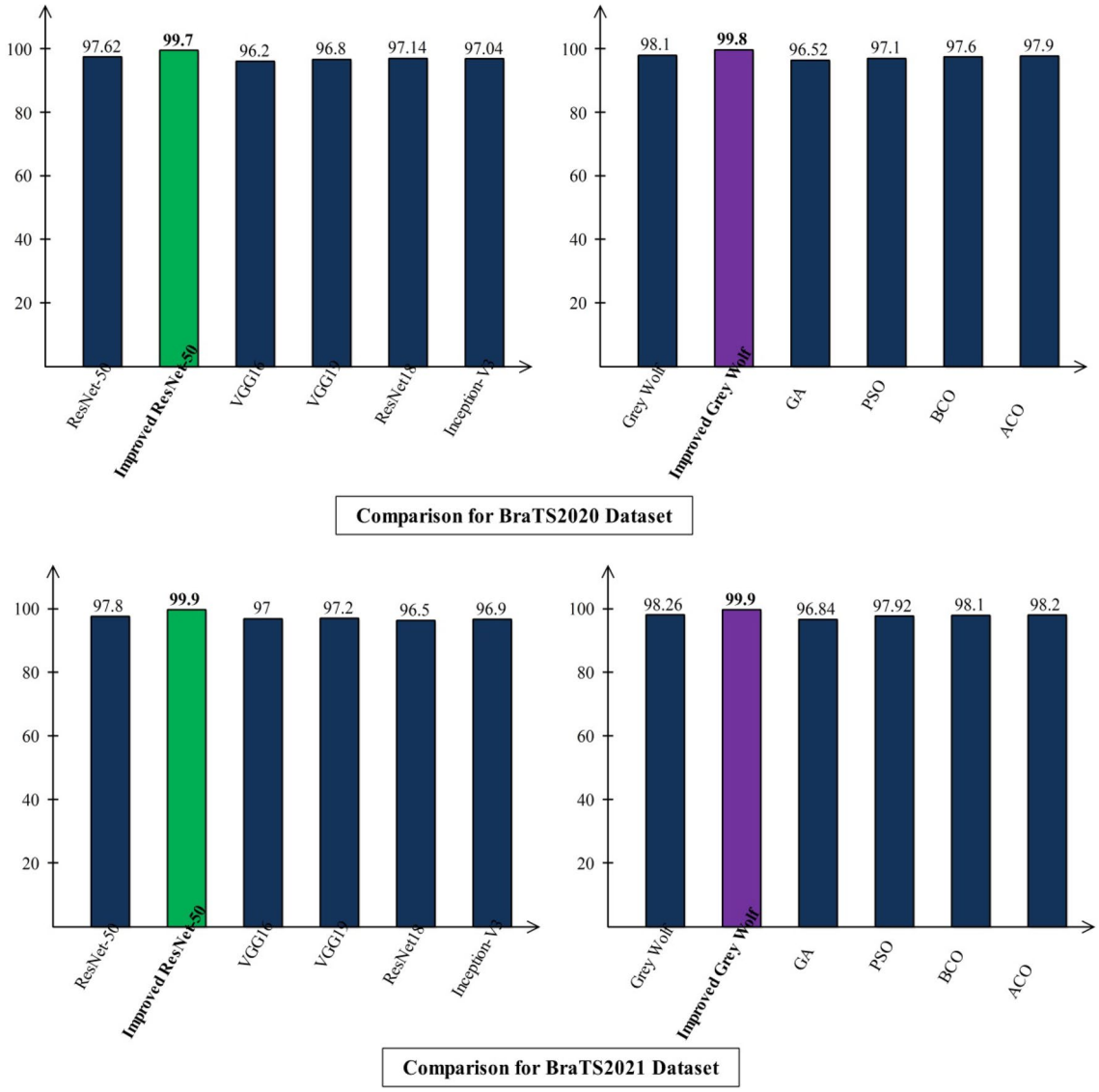

Figure 16. Comparison of original and improved CNN and optimization methods in terms of accuracy.

Reference	Year	Dataset	Accuracy (%)
Gull et al.[54]	2022	BraTS2020	98.25
Farajzadeh et al.[55]	2023	BraTS2020	98.93
Yadav et al.[56]	2023	BraTS2020	99.46
Ferdous et al.[57]	2023	BraTS2021	93.69
Proposed		BraTS2020	**99.80**
		BraTS2021	**99.90**

Table 12. Comparison of the proposed accuracy with SOTA techniques. Significant values are in bold.

In addition, a new Stacked Autoencoders network is designed and trained on the selected dataset. Features are extracted and fused using a novel fusion technique, showing improved accuracy. However, a few redundant features have been detected that were further removed using an improved version of the optimization technique. The experimental process is conducted on two publicly available datasets and obtained with improved accuracy. The improved version of the ResNet-50 model improved the classification accuracy and reduced the number of total parameters. In addition, the fusion process improved the accuracy, but time is increased. To address this challenge, a feature selection technique is proposed that maintained the accuracy and almost 70% reduced the computational time.

Data availability

The data that support the findings of this study are openly available at: https://www.med.upenn.edu/cbica/brats2020/ and https://www.med.upenn.edu/cbica/brats2021/.

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References

1. Pichaivel, M., Anbumani, G., Theivendren, P. & Gopal, M. An overview of brain tumor. *Brain Tumors* **1**, 1–10 (2022).
2. Kristensen, B., Priesterbach-Ackley, L., Petersen, J. & Wesseling, P. Molecular pathology of tumors of the central nervous system. *Ann. Oncol.* **30**, 1265–1278 (2019).
3. David, D. & Arun, L. Classification of brain tumor type and grade using MRI texture and shape in a machine learning scheme. *Artech J. Eff. Res. Eng. Technol* **1**, 57–63 (2020).
4. N. B. T. Society. *Brain Tumor Facts*. (2022). https://braintumor.org/brain-tumors/about-brain-tumors/brain-tumor-facts/
5. Boutry, J. *et al.* The evolution and ecology of benign tumors. *Biochim. Biophys. Acta Rev. Cancer* **1877**, 188643 (2022).
6. Jena, B. *et al.* Brain tumor characterization using radiogenomics in artificial intelligence framework. *Cancers* **14**, 4052 (2022).
7. Mehnatkesh, H., Jalali, S. M. J., Khosravi, A. & Nahavandi, S. An intelligent driven deep residual learning framework for brain tumor classification using MRI images. *Expert. Syst. Appl.* **213**, 119087 (2023).
8. Ostrom, Q. T., Cioffi, G., Waite, K., Kruchko, C. & Barnholtz-Sloan, J. S. CBTRUS statistical report: Primary brain and other central nervous system tumors diagnosed in the United States in 2014–2018. *Neuro-oncology* **23**, 1–105 (2021).
9. Demir, F., Akbulut, Y., Taşcı, B. & Demir, K. Improving brain tumor classification performance with an effective approach based on new deep learning model named 3ACL from 3D MRI data. *Biomed. Signal Process. Control* **81**, 104424 (2023).
10. Kurdi, S. Z. *et al.* Brain tumor classification using meta-heuristic optimized convolutional neural networks. *J. Pers. Med.* **13**, 181 (2023).
11. Wellington, S. L. & Vinegar, H. J. X-ray computerized tomography. *J. Petrol. Technol.* **39**, 885–898 (1987).
12. Rooney, W. *MRI: From Picture to Proton* (2003).
13. Jayadevappa, D., Srinivas Kumar, S. & Murty, D. Medical image segmentation algorithms using deformable models: A review. *IETE Tech. Rev.* **28**, 248–255 (2011).
14. Yazdani, S., Yusof, R., Karimian, A., Pashna, M. & Hematian, A. Image segmentation methods and applications in MRI brain images. *IETE Tech. Rev.* **32**, 413–427 (2015).
15. Whelan, H. T., Clanton, J. A., Wilson, R. E. & Tulipan, N. B. Comparison of CT and MRI brain tumor imaging using a canine glioma model. *Pediatr. Neurol.* **4**, 279–283 (1988).
16. Villanueva-Meyer, J. E., Mabray, M. C. & Cha, S. Current clinical brain tumor imaging. *Neurosurgery* **81**, 397–415 (2017).
17. Khan, M. A. *et al.* Multimodal brain tumor detection and classification using deep saliency map and improved dragonfly optimization algorithm. *Int. J. Imaging Syst. Technol.* **33**, 572–584 (2022).
18. Choy, G. *et al.* Current applications and future impact of machine learning in radiology. *Radiology* **288**, 318 (2018).
19. Mazurowski, M. A., Buda, M., Saha, A. & Bashir, M. R. Deep learning in radiology: An overview of the concepts and a survey of the state of the art with focus on MRI. *J. Magn. Reson. Imaging* **49**, 939–954 (2019).
20. Saba, L. *et al.* The present and future of deep learning in radiology. *Eur. J. Radiol.* **114**, 14–24 (2019).
21. Yin, S. *et al.* Development and validation of a deep-learning model for detecting brain metastases on 3D post-contrast MRI: a multi-center multi-reader evaluation study. *Neuro-oncology* **1**, 025 (2023).
22. Dequidt, P. *et al.* Exploring radiologic criteria for glioma grade classification on the BraTS dataset. *IRBM* **42**, 407–414 (2021).
23. Tandel, G. S. *et al.* Role of ensemble deep learning for brain tumor classification in multiple magnetic resonance imaging sequence data. *Diagnostics* **13**, 481 (2023).
24. Garg, G. & Garg, R. *Brain Tumor Detection and Classification Based on Hybrid Ensemble Classifier*. arXiv:2101.00216 (2021).
25. Chatterjee, S., Nizamani, F. A., Nürnberger, A. & Speck, O. Classification of brain tumours in MR images using deep spatiospatial models. *Sci. Rep.* **12**, 1505 (2022).
26. Hapsari, P. A. T. *et al.* Brain tumor classification in MRI images using en-CNN. *Int. J. Intell. Eng. Syst.* **14**, 437–451 (2021).
27. Yang, H., Ni, J., Gao, J., Han, Z. & Luan, T. A novel method for peanut variety identification and classification by improved VGG16. *Sci. Rep.* **11**, 1–17 (2021).
28. Amin, J., Sharif, M., Gul, N., Yasmin, M. & Shad, S. A. Brain tumor classification based on DWT fusion of MRI sequences using convolutional neural network. *Pattern Recogn. Lett.* **129**, 115–122 (2020).
29. Zahid, U. *et al.* BrainNet: Optimal deep learning feature fusion for brain tumor classification. *Comput. Intell. Neurosci.* **2022**, 1–10 (2022).
30. De Souza, D. L., Coelho, A. M., Felipe, J. C. & Baffa, M. D. F. O. Brain tumor classification on multi-modality mri using radiomic features. in *2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 1799–1804 (2021).
31. Gupta, T., Gandhi, T. K., Gupta, R. & Panigrahi, B. K. Classification of patients with tumor using MR FLAIR images. *Pattern Recogn. Lett.* **139**, 112–117 (2020).
32. Rajasree, R., Columbus, C. C. & Shilaja, C. Multiscale-based multimodal image classification of brain tumor using deep learning method. *Neural Comput. Appl.* **33**, 5543–5553 (2021).
33. Sharma, S. *et al.* Deep learning model for automatic classification and prediction of brain tumor. *J. Sens.* **2022**, 1–22 (2022).
34. Guo, S. *et al.* Multimodal MRI image decision fusion-based network for glioma classification. *Front. Oncol.* **12**, 1–10 (2022).
35. Fu, Y., Wu, X.-J. & Durrani, T. Image fusion based on generative adversarial network consistent with perception. *Inf. Fusion* **72**, 110–125 (2021).
36. Rehman, A. *et al.* Microscopic brain tumor detection and classification using 3D CNN and feature selection architecture. *Microsc. Res. Tech.* **84**, 133–149 (2021).
37. C. f. B. I. C. A. (CBICA). (2020). *Brain Tumor Segmentation (BraTS) Challenge 2020*. https://www.med.upenn.edu/cbica/brats2020/
38. C. f. B. I. C. A. (CBICA). (2021). *RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) Challenge 2021*. https://www.med.upenn.edu/cbica/brats2021/
39. Menze, B. H. *et al.* The multimodal brain tumor image segmentation benchmark (BRATS). *IEEE Trans. Med. Imaging* **34**, 1993–2024 (2015).
40. Bakas, S. et al. *Identifying the Best Machine Learning Algorithms for Brain Tumor Segmentation, Progression Assessment, and Overall Survival Prediction in the BRATS Challenge*. arXiv:1811.02629 (2018).
41. Bakas, S. *et al.* Advancing The Cancer Genome Atlas glioma MRI collections with expert segmentation labels and radiomic features. *Sci. Data* **4**, 170117 (2017).
42. Li, F.-F.I. Crowdsourcing, benchmarking & other cool things. *CMU VASC Semin.* **16**, 18–25 (2010).
43. He, K., Zhang, X., Ren, S. & Sun, J. Deep residual learning for image recognition. in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 770–778 (2016).

44. Theckedath, D. & Sedamkar, R. Detecting affect states using VGG16, ResNet50 and SE-ResNet50 networks. *SN Comput. Sci.* **1**, 1–7 (2020).
45. Ahsan, M. M. *et al.* Deep transfer learning approaches for Monkeypox disease diagnosis. *Expert Syst. Appl.* **216**, 119483 (2023).
46. da Silva, L. S. A., Lúcio, Y. L. S., Coelho, L. D. S., Mariani, V. C. & Rao, R. V. A comprehensive review on Jaya optimization algorithm. *Artif. Intell. Rev.* **56**, 4329–4361 (2023).
47. Yılmaz, E. K., Adem, K., Kılıçarslan, S. & Aydın, H. A. Classification of lemon quality using hybrid model based on Stacked AutoEncoder and convolutional neural network. *Eur. Food Res. Technol.* **1**, 1–13 (2023).
48. Chen, Y., Xia, R., Yang, K. & Zou, K. MFFN: Image super-resolution via multi-level features fusion network. *Vis. Comput.* **1**, 1–16 (2023).
49. Thakkar, A. & Lohiya, R. Fusion of statistical importance for feature selection in deep neural network-based intrusion detection system. *Inf. Fusion* **90**, 353–363 (2023).
50. Mirjalili, S., Mirjalili, S. M. & Lewis, A. Grey wolf optimizer. *Adv. Eng. Softw.* **69**, 46–61 (2014).
51. Saputra, D. C. E., Sunat, K. & Ratnaningsih, T. A new artificial intelligence approach using extreme learning machine as the potentially effective model to predict and analyze the diagnosis of anemia. *Healthcare* **1**, 697 (2023).
52. Rao, R. V., Savsani, V. J. & Vakharia, D. Teaching–learning-based optimization: A novel method for constrained mechanical design optimization problems. *Comput. Aided Des.* **43**, 303–315 (2011).
53. Di Leo, G. & Sardanelli, F. Statistical significance: p value, 0.05 threshold, and applications to radiomics—reasons for a conservative approach. *Eur. Radiol. Exp.* **4**, 18 (2020).
54. Gull, S., Akbar, S., Hassan, S. A., Rehman, A. & Sadad, T. Automated brain tumor segmentation and classification through MRI images. in *International Conference on Emerging Technology Trends in Internet of Things and Computing*, 182–194 (2021).
55. Farajzadeh, N., Sadeghzadeh, N. & Hashemzadeh, M. Brain tumor segmentation and classification on MRI via deep hybrid representation learning. *Expert Syst. Appl.* **224**, 119963 (2023).
56. Yadav, A. S. *et al.* A feature extraction using probabilistic neural network and BTFSC-net model with deep learning for brain tumor classification. *J. Imaging* **9**, 10 (2022).
57. Ferdous, G. J., Sathi, K. A., Hossain, M. A., Hoque, M. M. & Dewan, M. A. A. LCDEiT: A linear complexity data-efficient image transformer for MRI brain tumor classification. *IEEE Access* **11**, 20337–20350 (2023).

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Author contributions

Muhammad Sami Ullah and Muhammad Attique Khan provides the conceptualization, software requirement, contribututed in methodology and wrote the first draft of the manuscirpt. Nouf Abdullah Almujally and Majed Alhaisoni provides the Data curation, contributed in the methodology and investigated the results. Tallha Akram and Mohammad Shabaz provides the figures, contributed in the methodology and reviewed the final draft of the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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